

Azure Virtual Desktop: Real-World Use Cases

Case studies and practical applications
of AVD in various industries

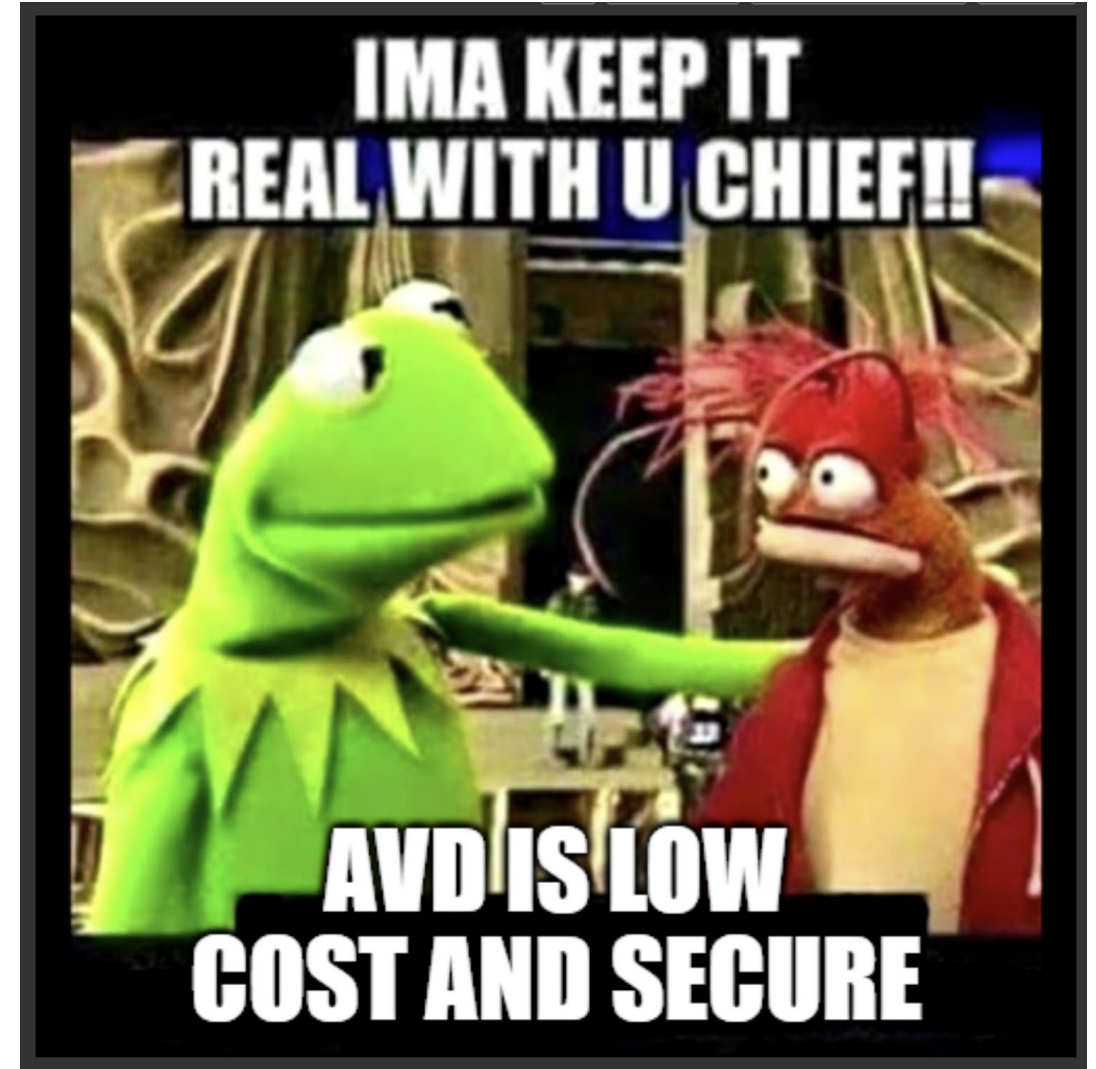
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A Few AVD Real World Use Cases to Consider

- Temporary / External Employees
(Intern season is approaching!)
- Cross Agency Access
 - Cross Cloud / Cross Tenant Registration
 - B2B
- Secure Network Access
- Power of Proximity
 - Desktops close to central data assets
 - Desktops in Remote Location
(near remote data!)



Azure Virtual Desktop

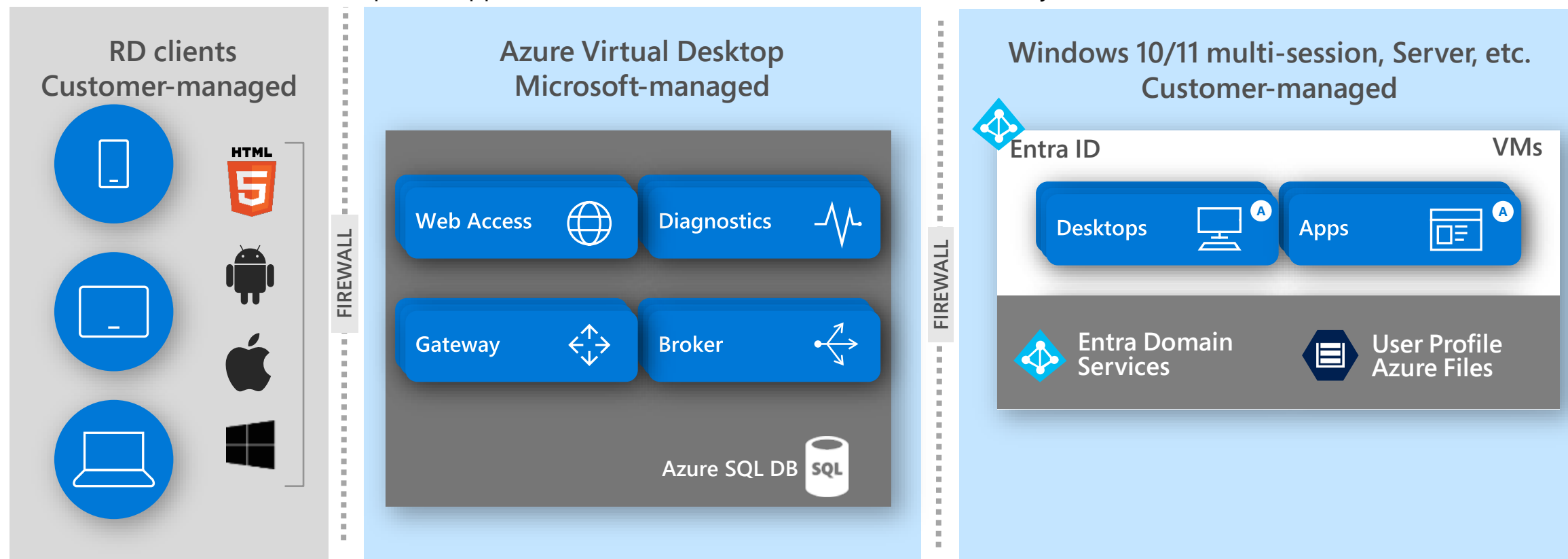
Consists of: **Azure service** to manage connections between **RD clients** and **Windows VMs** in Azure (e.g. Windows multi-session)

IT admins can:

- Publish remote desktops and apps to end users from pools of single or multi-session Windows VMs in Azure
- Manage and troubleshoot connections between RD Clients and Windows virtual machines

End users can:

- Connect to Windows desktops and applications from their favorite client device from anywhere on the internet

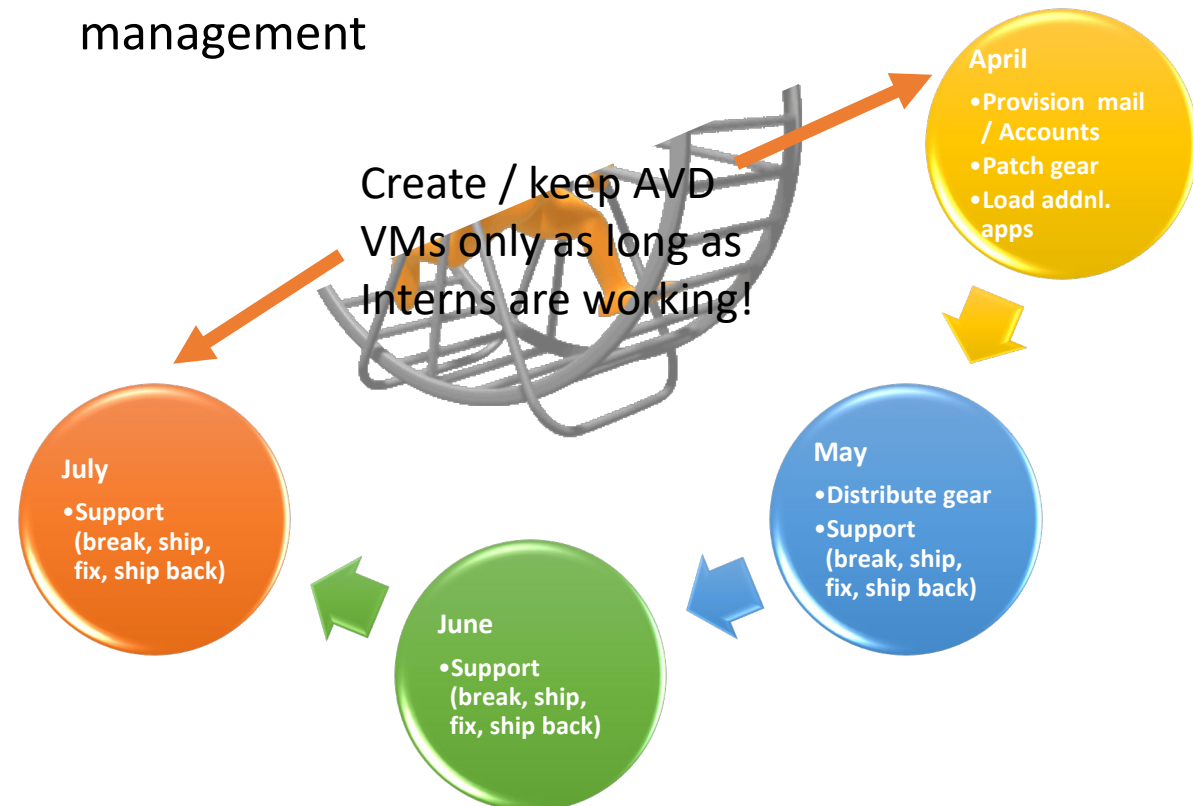


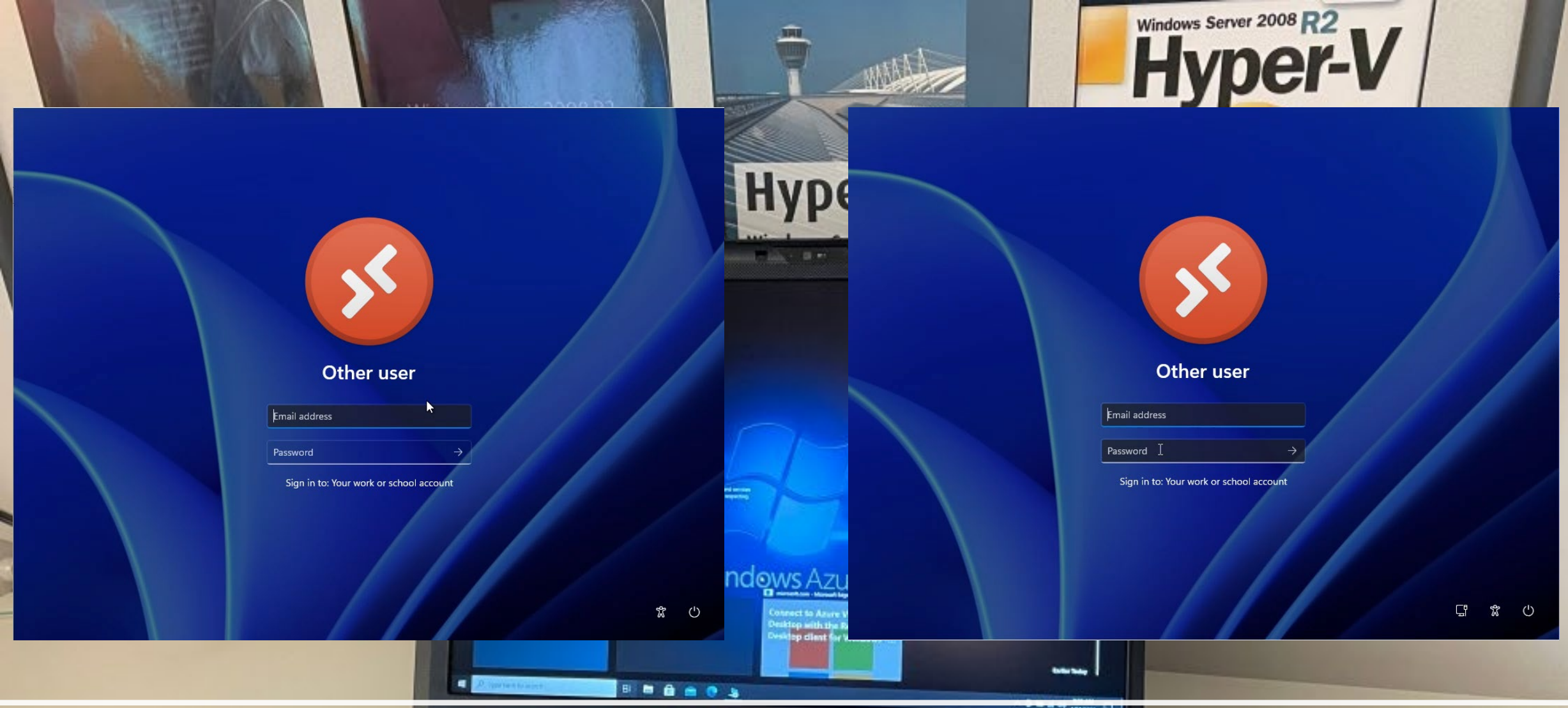
Interns (contractors, seasonal staff, “disaster teams”, temp call center) highlight cost / complexities of technology support in the post-COVID “work from anywhere” world

Why do you want to live on the gerbil wheel of hardware management...
for users who already (likely) have a laptop!

Deploy Azure Virtual Desktop!

- **Cut planning / deployment / support cycle in half**
- Leverage the laptop / desktop interns already have (or use old relics the agency has in a closet – it’s only going to be a terminal anyway!)
- **No “GFE” procurement / management**
- Use the internet they already have at home / school
- Wipe out hardware cost / burden of hardware management





Repurpose Existing Hardware? ... GitHub Repo for Kiosk

Auto-Start AVD Session Using URI Schemes – Azure Advanced Migration

External Users?

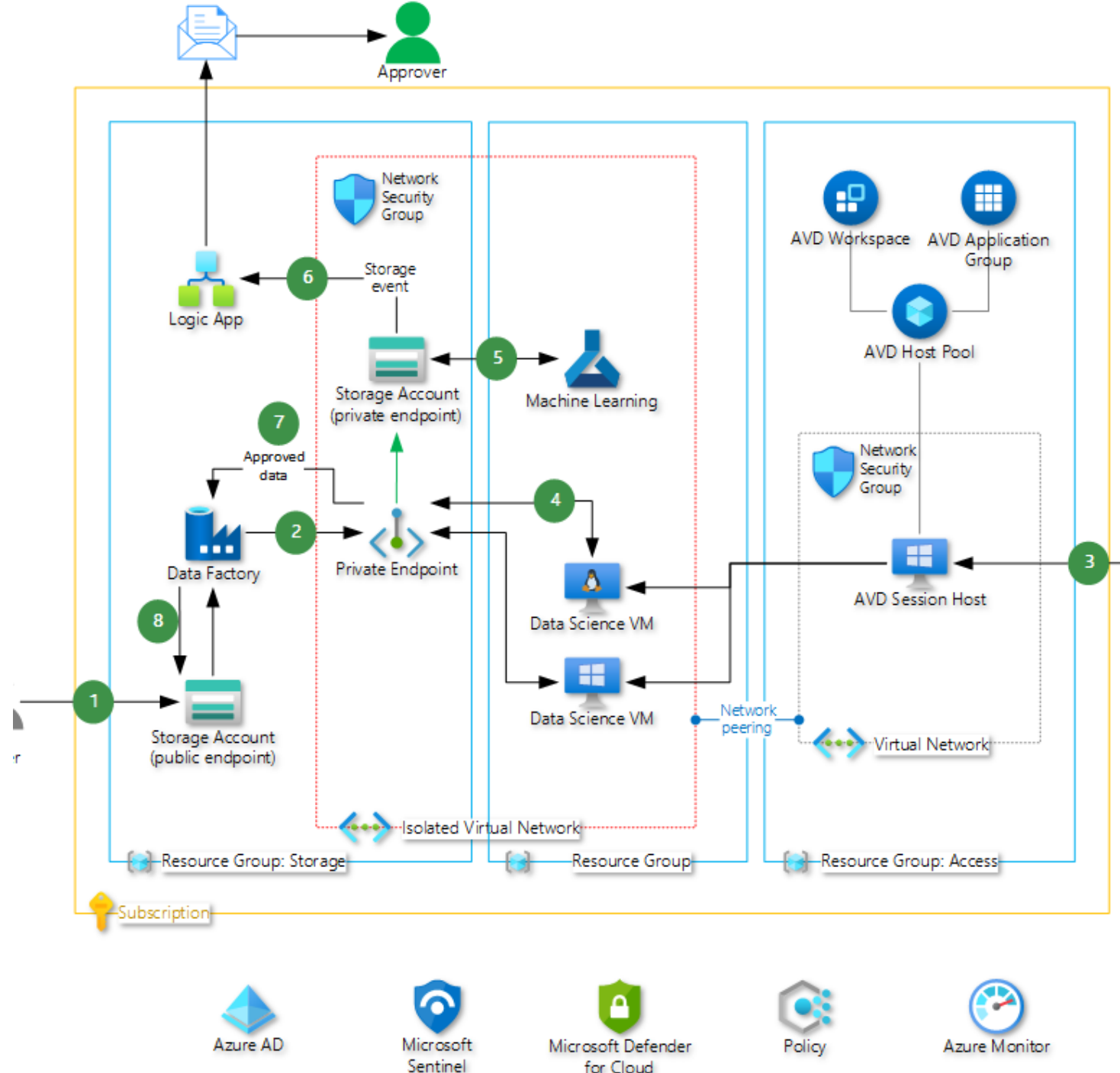
Non-Employee Access?

- ISVs sometime host their applications in Azure and publish via AVD using [Per-user Access for Azure Virtual Desktop](#)
 - Enables controlled access to applications and data
 - Published using Entra ID (can use smart card / MFA if desired)
 - Requires an Entra identity in the hosted tenant
- Government agencies have done the same thing:
 - **RemoteApp publishing for child welfare applications** controlled access to heritage app by non-employees
 - Contractor “sand box”
Remote access to your HVAC!

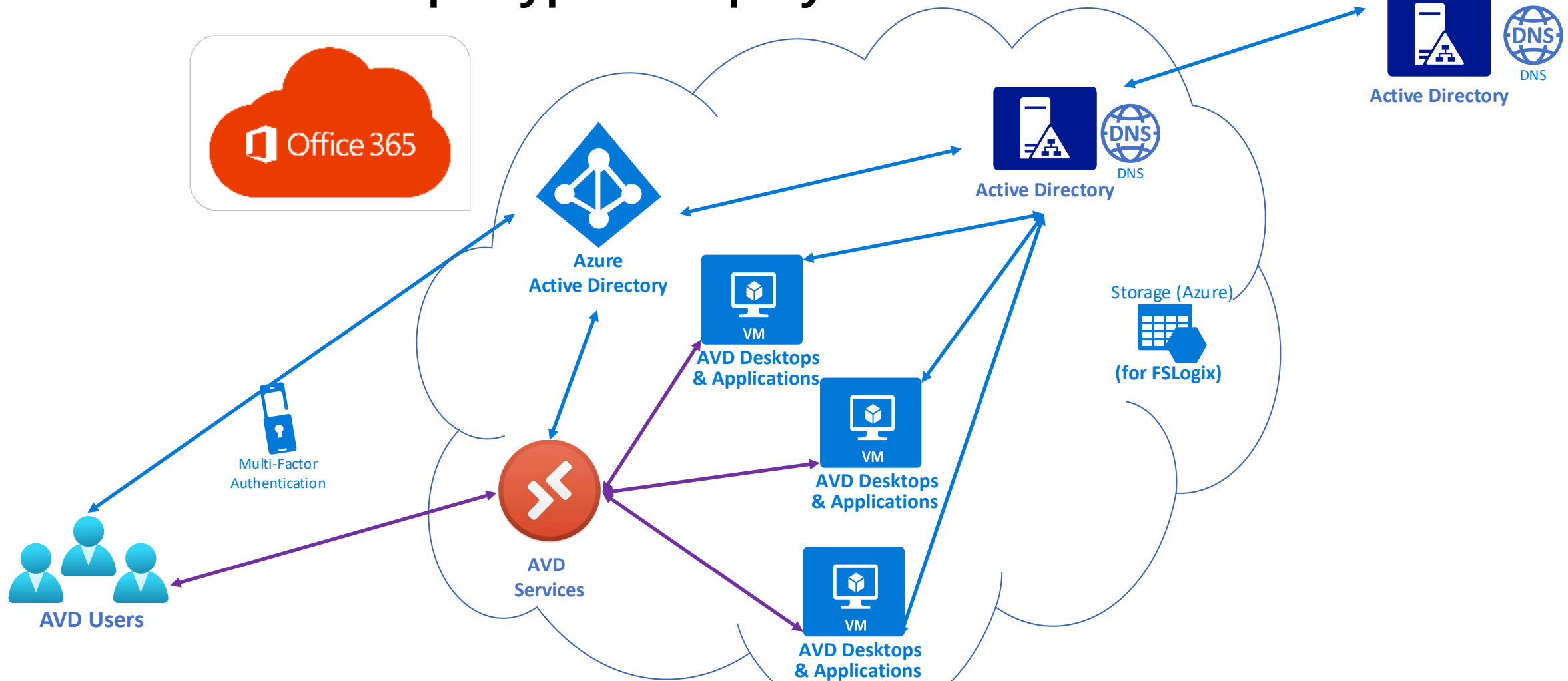


Secure Enclaves for Restricted Data?

- The [Secure Enclave for Research](#) (also known as the Secure Research Enclave) is a reference architecture for a remotely-accessible environment for researchers to use in a secure manner while working on restricted data sets. The solution features robust mechanisms for control over user access to the environment and also over movement of data in or out of scope for and so it is ideal for working with restricted data sets. Data in the environment can be analyzed with traditional VMs using Windows or Linux with well-known tools such as R Studio and supports the use of advanced analytical tools such as Azure Machine Learning.
- The solution is built using multiple Azure services including [Azure Virtual Desktop](#), Azure Key Vault, and Azure Data Factory to provide strong control over data movement into and out of the environment to prevent unauthorized exfiltration of data sets.
- This solution was created in collaboration with the University of Pittsburgh.



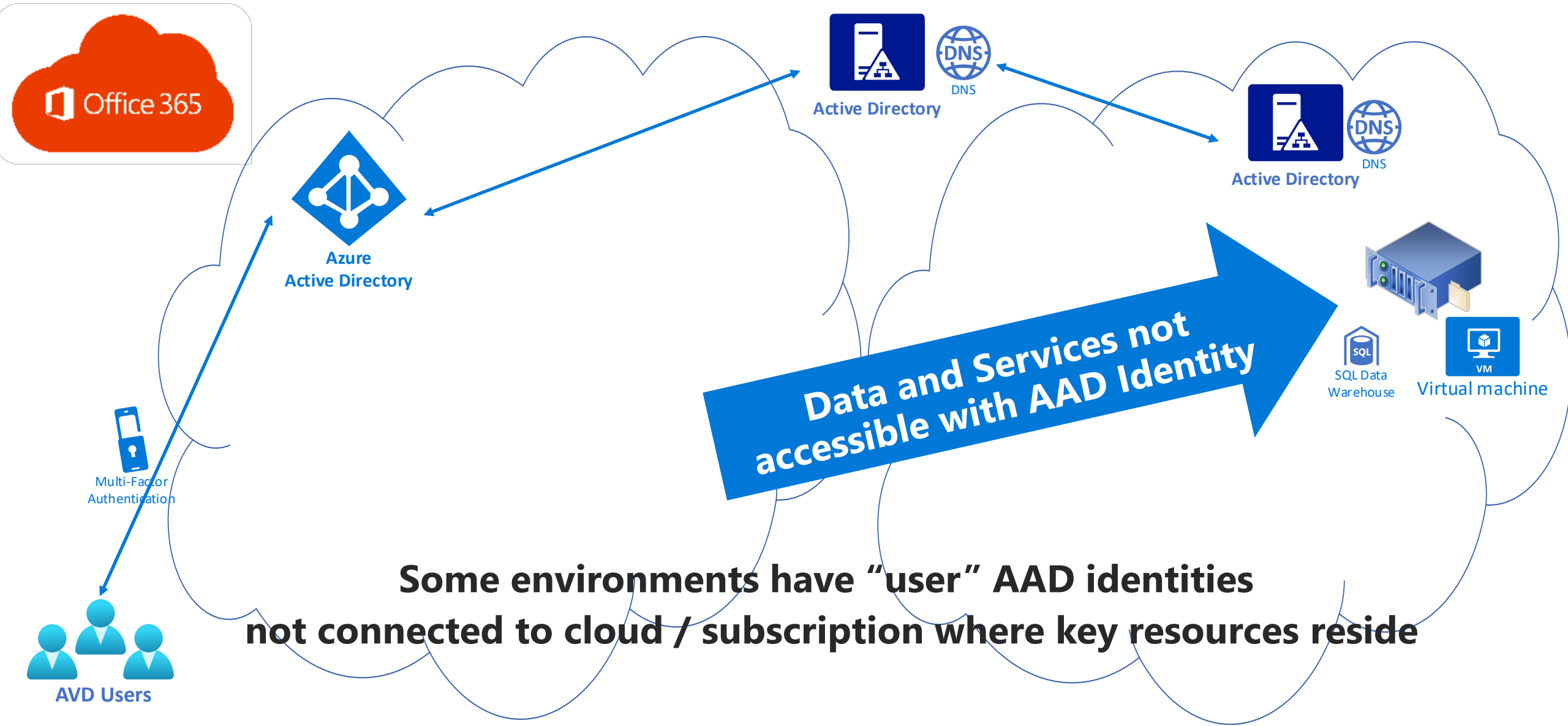
Azure Virtual Desktop: Typical Deployment



Typical Deployment is in the same cloud using AAD connected to subscription(s)

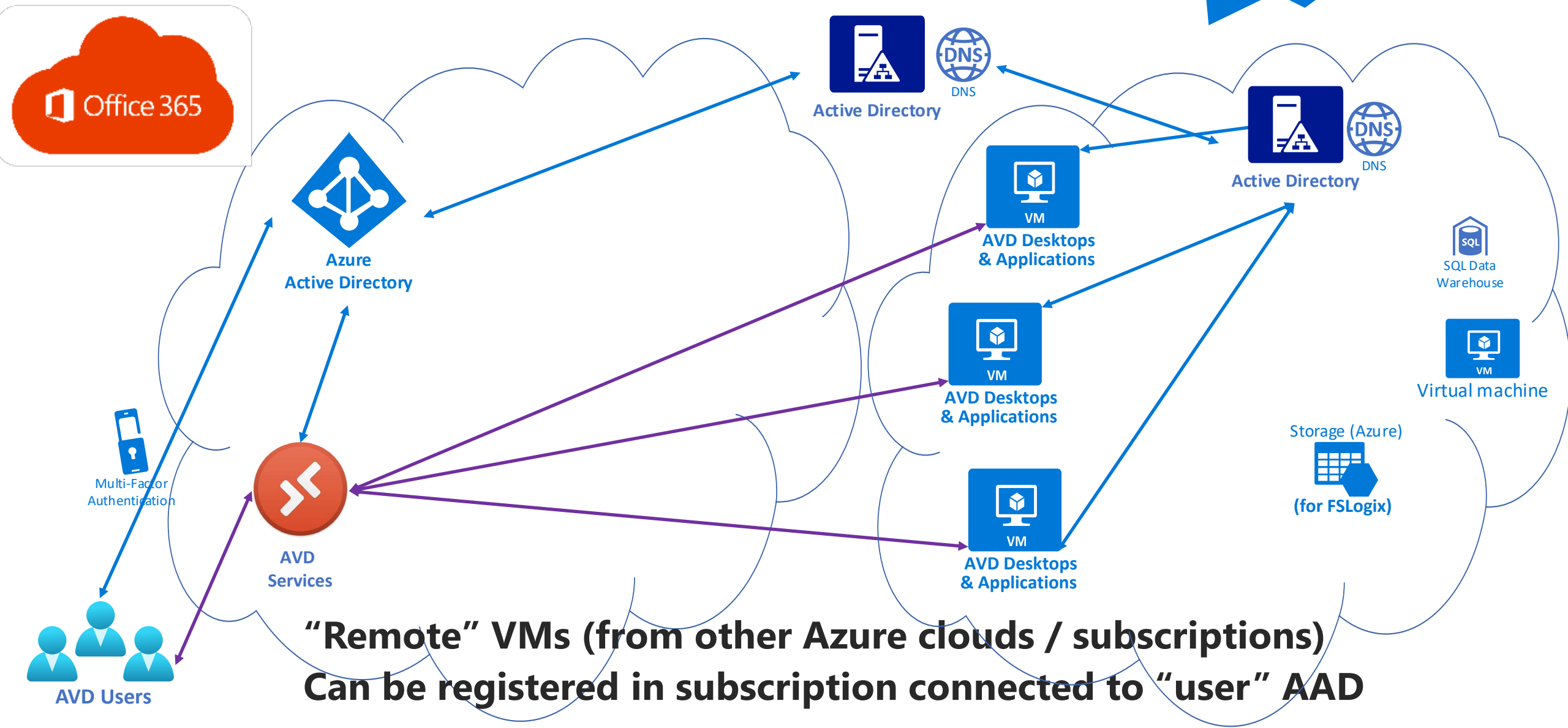
[Gov Topics: AVD Cross-Cloud / Cross-Tenant Deployment - Microsoft Tech Community](#)

Multiple Clouds (AADs / subscriptions) Makes Things Complicated



Azure Virtual Desktop: Cross-Cloud Registration

[BLOG POST: AVD Cross-Cloud / Cross-Tenant Deployment](#)



**"Remote" VMs (from other Azure clouds / subscriptions)
Can be registered in subscription connected to "user" AAD**



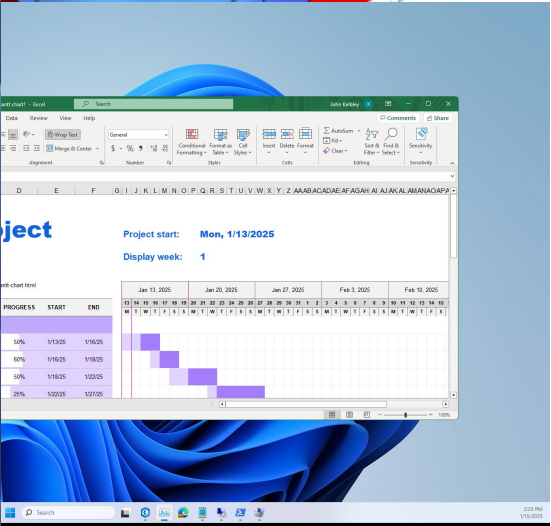
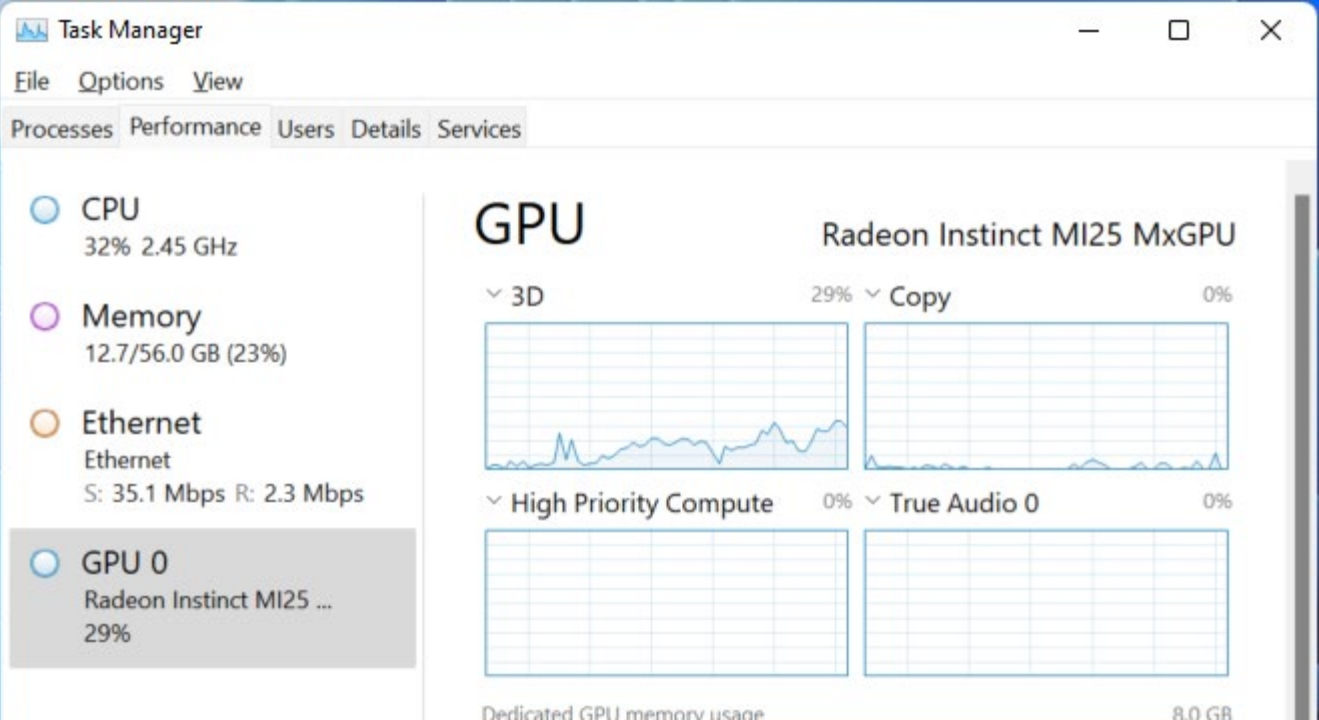
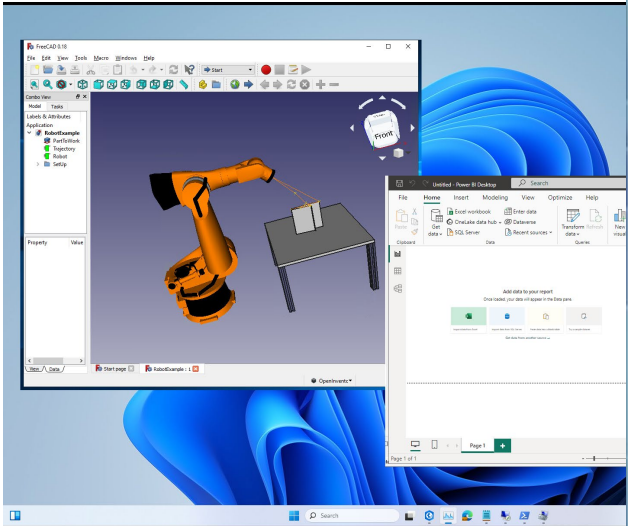
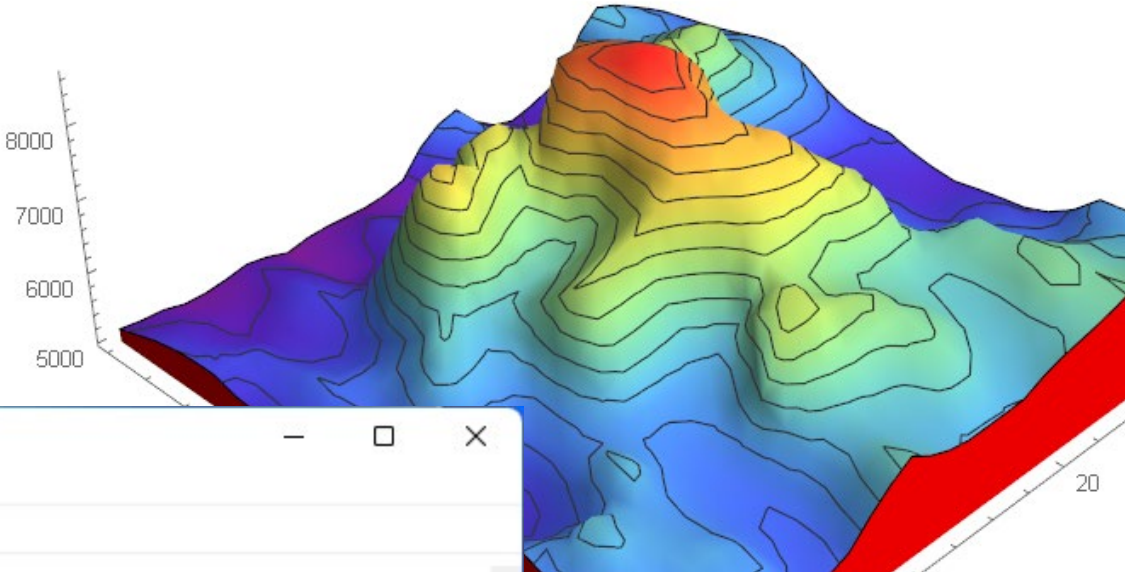
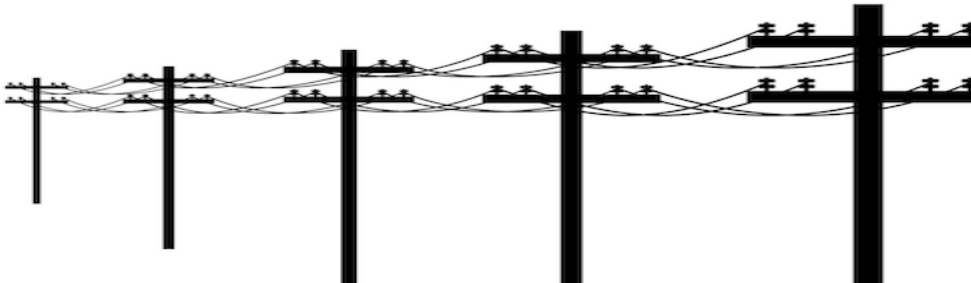
B2B for Enterprise Collaboration is Coming for AVD

- B2B enables **inviting guest users** and providing them access to resources **without giving them an agency managed identity**
- The guest users **use their own account to access virtual apps and desktops**
- Lowers management cost
- B2B will allow ISVs to invite end-users from different companies to the ISV's tenant, provide user separation and limit the resources they have access to
- B2B also lowers management costs from things like password reset

Data Analytics is Best NEAR the Data!

Distance = Latency!

Mountains of Data



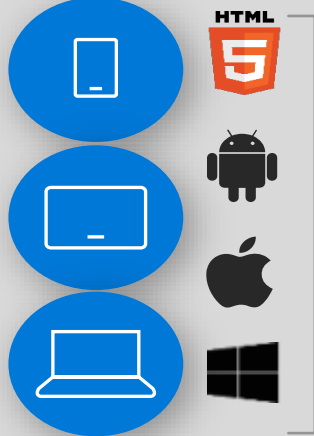
High-Level Architecture: AVD Compared to Azure Local

Azure Virtual Desktop

**Azure
Public /
Azure Gov**

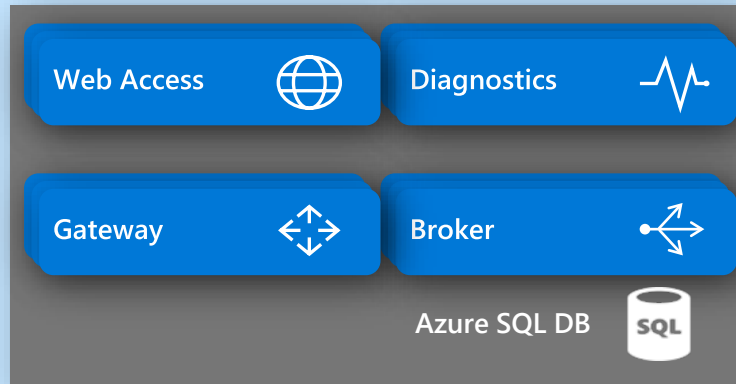
User experience
(typically)
**delivered over
Internet**

RD clients
Customer-managed



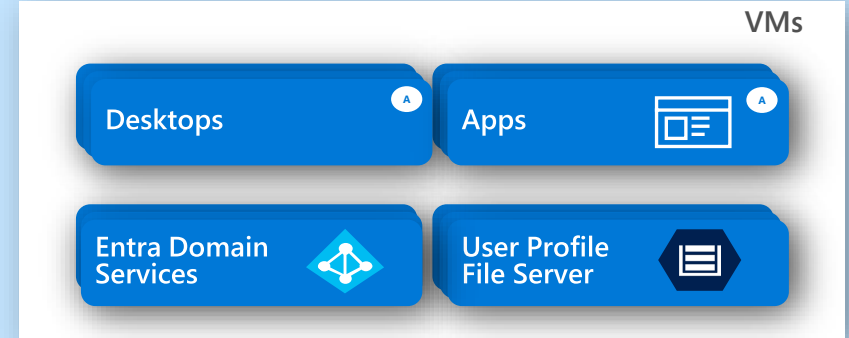
FIREWALL

Azure Virtual Desktop
Microsoft-managed



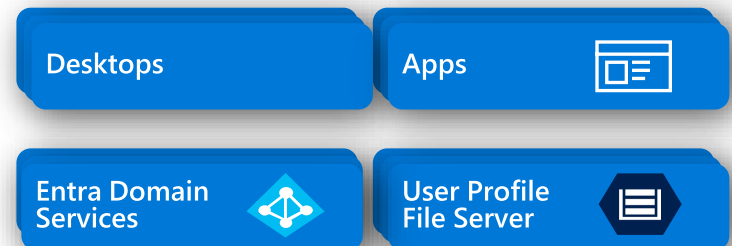
FIREWALL

Windows 7 / 10 / 11 single or 10 / 11 multi-session, Server
Customer-managed



Express Route / S2S VPN to on premises

(on Azure Local)



**AVD VMs run inside
customer Azure subscription
Gateway /**

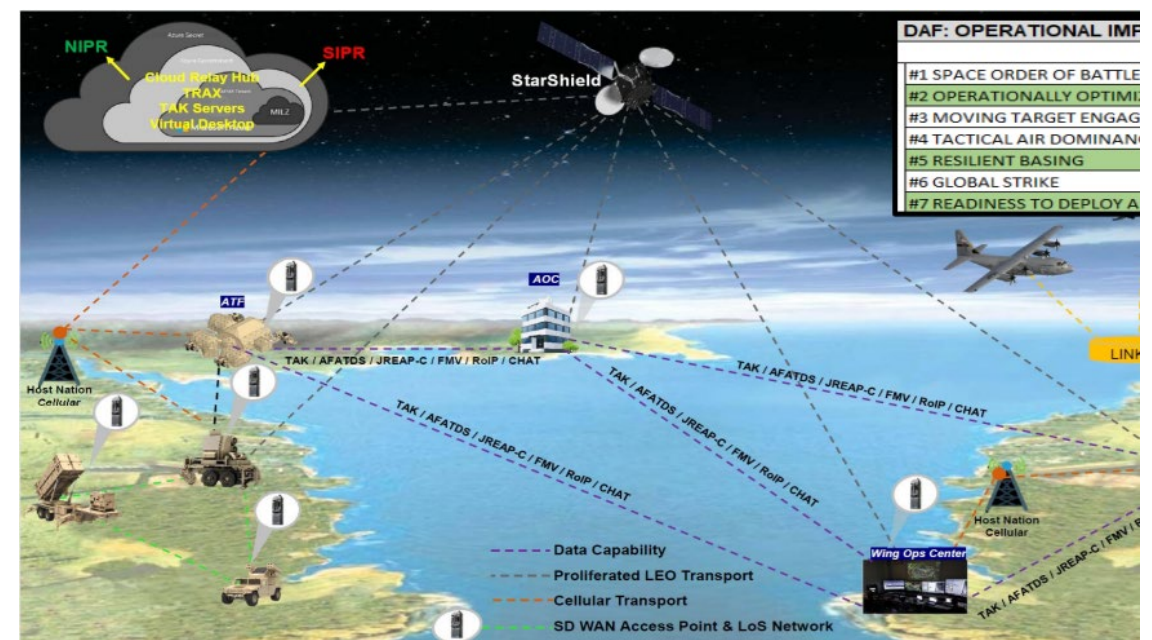
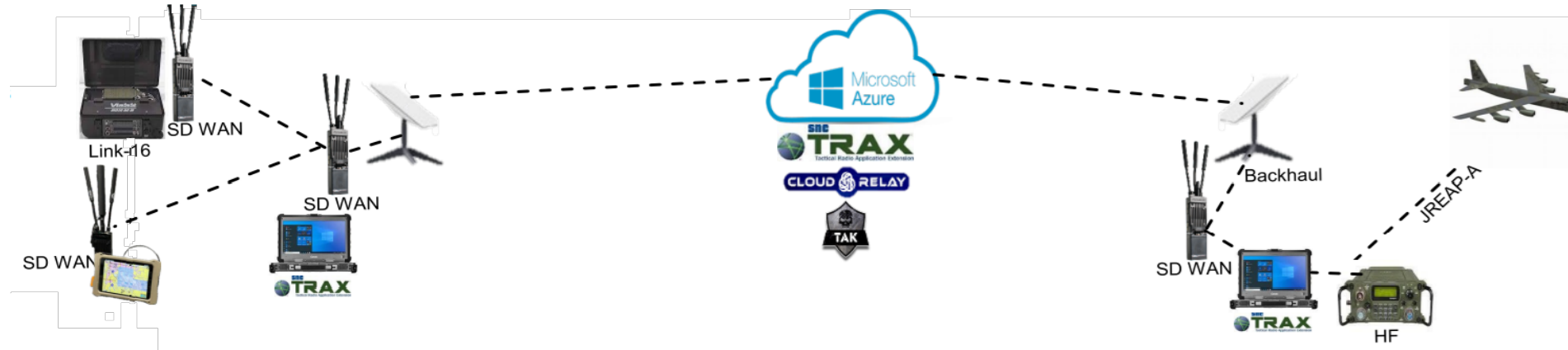
RDP ShortPath for Public Networks
uses best "available" route
after session started

(low latency!)

What is MCHMR?



- Manet C2 High Mobility Radio (MCHMR)
- Uses existing DoD investments to connect geographically separated units throughout the theater through Software-Defined Network Mesh & Azure infrastructure
- Empower mission owners with an agile “fight tonight” operational capability





MCHMR Compute Package Components

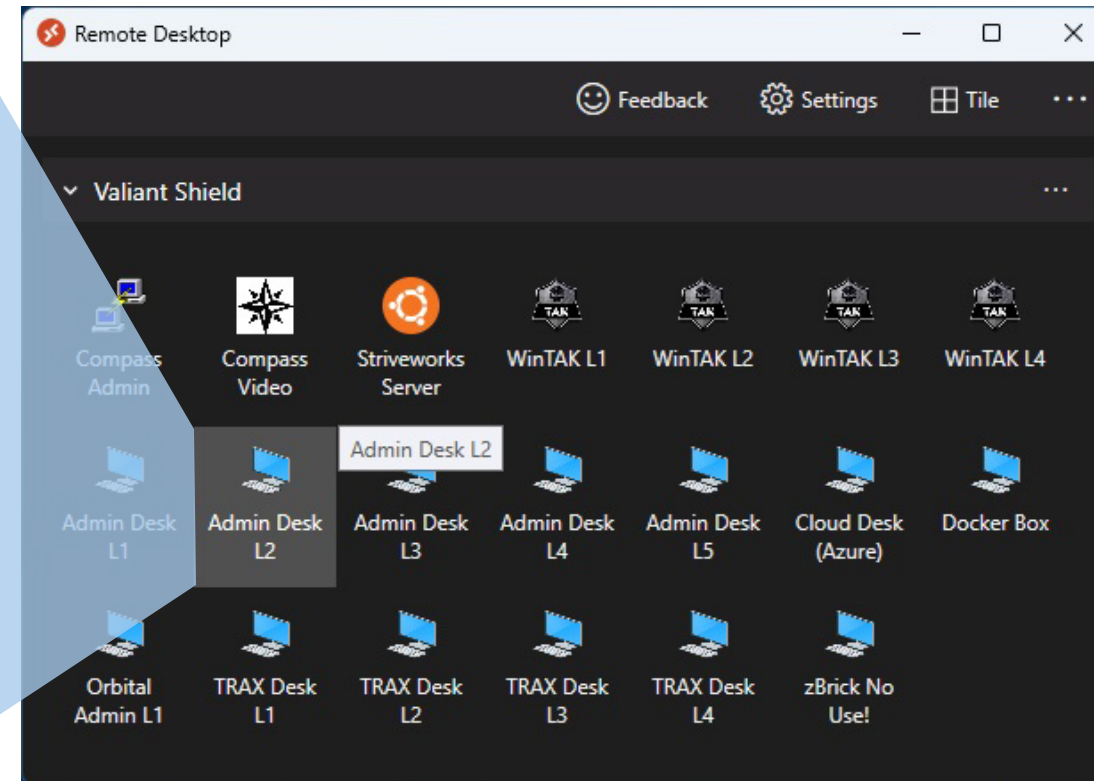
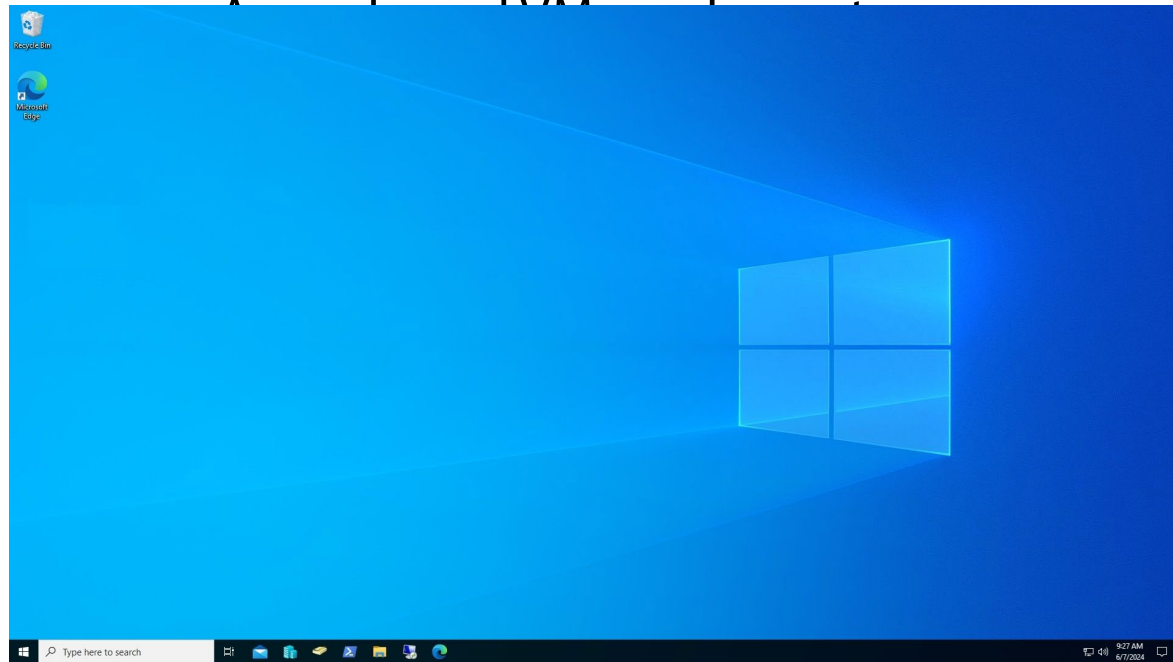
- 4U package (a "U" is a 1.75 inch computer blade)
- 1 Network switch
- 3 servers (Dell XR4510c)
- 1 MPU5 Mesh Radio
- 2 110v Power cords
- 5+ network cables and related connectors (SFPs and other)



Azure Virtual Desktop for Access



- AVD allowed for secure access to





Demonstration: Azure Virtual Desktop on Azure Stack HCI

John Kelbley

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Thank You!

Learn more about Azure Virtual Desktop – Additional sessions Today!

- 10:30AM – 11:00AM Introduction to Azure Virtual Desktop
- 12:45PM – 1:30PM Azure Virtual Desktop: Real World Use Cases
- **1:45PM – 2:30PM** **Deploying and Managing AVD**
- 3:00PM – 3:45PM Azure Arc: Optimizing Server Management
- 5:30PM – 6:30PM Happy Hour sponsored by 

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FEDERAL AZURE SUMMIT AGENDA

JAN 16 2025

8:30 - 9:00

Registration and Welcome Coffee

9:00 - 9:15

Opening Remarks by [Javier Vasquez](#)

9:15 - 9:45

Keynote: Future of Hybrid Cloud by [Jason Payne](#)

9:45 - 10:15

Adaptive Cloud: Extending Cloud Capabilities to the Edge by [Eric Sanchez](#) / [Joe Fox](#)

10:15 - 10:30

Morning Break

10:30 - 11:00

Introduction to Azure Virtual Desktop (AVD) by [John Kelbley](#) / [Shawn Meyer](#)

11:00 - 11:45

Introduction to Azure VMWare Solution by [Ruth Maygar](#) / [Kevin Jellow](#) / [Connor Loving](#)

11:45 - 12:45

Lunch Sponsored by



12:45 - 1:30

[Session 1A: Real-World Use Cases of Azure Virtual Desktop](#) by [John Kelbley](#) / [Shawn Meyer](#)

[Session 1B: Real-World Use Cases of Azure Edge Devices](#) by [Rich Shillington](#) / [Jason Benson](#)

1:45 - 2:30

[Session 2A: Deploying and Managing AVD](#) by [Shawn Meyer](#)

[Session 2B: Real-World Use Cases of Azure VMWare Solutions](#)
by [Ruth Maygar](#) / [Kevin Jellow](#) / [Connor Loving](#)

2:30 - 2:45

Afternoon Break

3:00 - 3:45

[Session 3A: Azure Arc – Optimizing Server Management](#) by [John Kelbley](#)

[Session 3B: Deploying & Managing Azure VMWare Solutions](#)
by [Ruth Maygar](#) / [Kevin Jellow](#) / [Connor Loving](#)

4:00 - 4:45

Fed Chief Architect Panel: The Future of Cloud and Edge Computing by [Jason Payne](#) / [Chief Architects](#)

4:45 - 5:00

Closing Remarks and Networking by [Javier Vasquez](#)

5:30 - 6:30

Happy Hour Sponsored by



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The benefits of Nerdio



Cost insights &
reporting



Image lifecycle
management



Environment
rightsizing



DR & capacity
mitigation



Application
management



Policy
management



HCI stack
support



Modern endpoint
management



End-user self
service